29 October 2026

The Editors
PLOS ONE

Re: Research Article submission, "Allowance edges or transfer edges? Evaluating on-chain smart contract reputation out of window against degree baselines"
Running title: *Allowance versus transfer edges for smart contract reputation*

Dear Editors,

We submit the above manuscript for consideration as a Research Article in *PLOS ONE*. Before a user trades through a router or deposits into a vault, the user signs an ERC-20 approval that lets that smart contract move the user's tokens. The manuscript asks what these approvals, recorded on chain as allowances, add to graph reputation scores for smart contracts compared with the token transfers on which existing scores are built. On public data from the Arbitrum One rollup we compare PageRank on the latest allowances (EndorseRank) with the published transfer-based Adaptive Weighted PageRank (AWP) and with the raw degree counts that both walks smooth, for 6,483 contracts, and we freeze every score before the outcomes it is tested on.

Why the work matters beyond its case study. Wallets, protocols and security tools must decide which contracts deserve spending authority, and graph reputation scores are one proposed aid. Our results show why such scores should be tested out of window and against the counts they smooth. Neither walk beat the plain count of its own edges in two windows: the transfer walk fell below the count of distinct senders, and the allowance walk below the count of approving owners, in comparisons made after the labels were known; a closed form shows that for most contracts the allowance walk is a reweighted approval count. On an outcome that neither graph contains, the later liquidation avoidance of contracts trading on a perpetual futures exchange, the approval count ranked traders above the transfer count in the primary window when the rule registered before this study extracted the July–September outcomes was applied to the revised scores, though not in the registered analysis; the PageRank versions were not resolved in either. The evaluation design and the code are public.

Why *PLOS ONE*. The journal assesses scientific soundness rather than perceived impact and publishes null and negative results. Several of our central findings are negative or qualified: no PageRank beat the raw count of its edges, whatever its restart rule; the PageRank comparison on trader outcomes was not resolved; and the one positive trader result, a lead of the approval count, holds in the re-analysis but not in the registered analysis. We also report a deviation openly: after the results of the registered analysis were known we defined the analysis we report, with token amounts valued in US dollars over five widely traded tokens and every score computed from the twelve months before its freeze. The manuscript reports this revised analysis, applies the registered rules to it as a re-analysis, sets the registered results beside it, and gives the plan and the registration verbatim, with their commits and hashes, as supporting information. Every effect size comes with its paired bootstrap interval.

Statements. The manuscript is not under consideration elsewhere and has not been published. It is derived from the first author's doctoral research at the University of South Africa (UNISA). A related manuscript by the same authors, on a walk that mixes the two edge types, the decision rules fixed for it and the Sybil exposure of the allowance walk, is being submitted at the same time to *IEEE Access*; we upload it as a related manuscript file. The two papers share the data, the cohorts, the registration and the revised analysis, and both report the spender holdout coefficients of the raw degrees, EndorseRank and AWP with their degree contrasts, the closed form of EndorseRank and the solver runtimes. The allowance-against-transfer comparison on trader outcomes is reported only in this manuscript; on the same trader cohort the related manuscript reports only the contrasts of its mixed walk that its own registered rule names. The code, configuration, queries, result summaries, per-contract scores and labels, and pre-specification documents are publicly available at `https://github.com/abitocodes/ERC20-Allowance-PageRank-Wallet-Reputation` and `https://github.com/abitocodes/phd-dissertation-draft-en`. The study uses only public blockchain data and public token prices and involves no human participants; only contracts are scored. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors received no specific funding for this work. Claude (Anthropic) was used to write and run analysis code, generate tables and figures, and draft and revise the text; the full statement is in Methods (Use of generative AI).

Thank you for considering our work.

Yours sincerely,

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